

Extrauterine Pregnancy 11. Emergencies, Differential Diagnosis, and Systems

Companion reading handout for simultaneous listening.

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Transcript

Extrauterine Pregnancy audio course. Episode 11. Emergencies, Differential Diagnosis, and Systems.

The emergency lesson in ectopic pregnancy is that diagnosis and resuscitation do not always proceed at the same speed. A stable patient with a nondiagnostic scan can be followed carefully. A patient with shock, peritonism, syncope, or increasing hemoperitoneum needs escalation before the diagnostic label is academically perfect.

ACOG states that tubal ectopic pregnancy in an unstable patient is a medical emergency requiring prompt surgical intervention. The same bulletin tells clinicians to evaluate and treat urgently when there are signs of rupture, hemodynamic instability, or an acute abdomen. That language is deliberately simple because delay is the preventable harm. The danger is not only that a tube ruptures. It is that clinicians wait for a laboratory value, a repeat scan, or a more satisfying diagnostic sentence while the patient is bleeding.

The listener should let this principle override the rest of the course when necessary. Most of this course teaches diagnostic precision because stable patients can be harmed by premature treatment. Emergency care is the opposite moment. When the patient is physiologically declaring hemorrhage, waiting for perfect diagnostic precision becomes the harm. The skill is knowing which mode you are in: stable uncertainty or unstable bleeding.

Novak's description of rupture is clinically useful because it separates early nonspecific findings from decompensation. Before rupture, vital signs can be normal; abdominal tenderness may be mild; the uterus may feel only slightly enlarged; adnexal tenderness or mass can be absent or misleading. With rupture and intra-abdominal hemorrhage, tachycardia is followed by hypotension, bowel sounds decrease, the abdomen becomes distended and tender, rebound may appear, and pelvic examination may become limited by pain and guarding. The sequence matters: hypotension can be late.

Repeat that last sentence because it is easy to forget during a busy shift. Hypotension can be late. Tachycardia, pallor, syncope, worsening pain, guarding, and ultrasound blood may be earlier. If the resident waits for low blood pressure before believing hemorrhage, the system may start moving only after compensation has failed.

A normal first hemoglobin also does not make acute bleeding safe. In early intra-abdominal hemorrhage, measured hemoglobin may lag behind blood loss until redistribution and fluids reveal the deficit. The resident should respond to physiology: syncope, pallor, worsening pain, peritoneal signs, rising pulse, falling pressure, ultrasound blood, and clinical appearance. A number that has not yet fallen cannot overrule an abdomen that is declaring hemorrhage.

Keep three unreliable reassurances together: normal blood pressure, normal first hemoglobin, and low hCG. Normal blood pressure may only mean compensation is still working. Normal hemoglobin may only mean the blood loss is too acute to have equilibrated in the measured sample. Low hCG may only mean a small amount of trophoblast, not a safe location. None of these findings excludes rupture. In a concerning patient, they are data points to interpret, not permission to stand down.

The same principle applies to pain medication and temporary improvement. A patient may have less pain after analgesia while still bleeding. She may have a moment of less tubal stretch after rupture and still have hemoperitoneum. She may look calmer because she is lying still and compensating. Reassessment should ask whether the physiology is improving, not only whether the patient looks quieter.

Ultrasound free fluid should be interpreted by amount, echogenicity, distribution, and clinical context. Farren emphasizes the need to distinguish rupture with ongoing major hemorrhage from smaller self-limited blood loss through the fimbrial end of an intact tube. Fluid limited to the pelvis is not the same as blood reaching the upper abdomen. Morrison pouch fluid in a supine patient indicates severe hemoperitoneum, with Farren citing a threshold above 670 milliliters in that setting. Right upper quadrant scanning can therefore speed triage in an unwell patient.

The Morrison pouch number should be remembered as a severity signal, not as a calculation to perform at the bedside. If blood has reached the hepatorenal space in a supine patient, the pelvis has overflowed. That means the scan is showing a volume and distribution of bleeding that should accelerate surgical and resuscitation planning. Upper abdominal fluid turns pelvic pain into a whole-abdomen hemorrhage problem.

ACEP's emergency medicine policy adds another safety rule: do not use hCG alone to exclude ectopic pregnancy. Symptomatic early pregnancy deserves pelvic ultrasound regardless of whether hCG is below a traditional discriminatory threshold. This is the emergency-room translation of the same principle taught earlier in the course. hCG describes trophoblast activity over time; it does not prove location and does not protect the patient from rupture at low values.

When the patient is concerning, the system should move before collapse. Establish large-bore intravenous access. Send complete blood count, coagulation studies when indicated, type and screen or crossmatch, and prepare blood. Involve senior gynecology, anesthesia, the operating room, and blood bank early. Continue ultrasound only if it helps immediate decisions and does not delay source control. If the patient is unstable, the question becomes how to control hemorrhage, not how to produce the most elegant diagnostic pathway.

Notice the order of work here. Access, blood, anesthesia, operating room, and senior help are parallel actions. They are not sequential tasks to begin after the next scan image. In hemorrhage, time is lost when each person waits for another person to become convinced. The resident's job is to make the concern explicit early enough that the system can run in parallel.

This is why early calling for help is a clinical skill rather than an administrative habit. Ectopic hemorrhage requires several parallel processes: vascular access, blood availability, anesthesia readiness, surgical preparation, informed consent when possible, and transport. A single resident quietly repeating vital signs cannot create those processes after the patient has crashed. The system must be activated while compensation is still present.

The practical division of labor should be explicit. One clinician names the concern and keeps reassessing the patient. One obtains access, blood tests, type and screen or crossmatch, and starts fluid or blood as indicated. One calls anesthesia, the operating room, blood bank, and senior surgical help. One keeps the patient and family informed when time allows. Emergencies fail when all of these tasks wait behind a single unfinished diagnostic step.

Differential diagnosis still matters. Pregnancy plus abdominal pain is not synonymous with ectopic pregnancy. Early pregnancy loss, ruptured corpus luteum, ovarian torsion, pelvic inflammatory disease, tubo-ovarian abscess, appendicitis, renal colic, pyelonephritis, hepatobiliary disease, gastrointestinal disease, trauma, adnexal mass, and gestational trophoblastic disease can all overlap with parts of the presentation. Novak specifically reminds the reader that adnexal masses can represent corpus luteum, endometrioma, hydrosalpinx, ovarian neoplasm, dermoid cyst, or pedunculated fibroid rather than ectopic pregnancy.

The differential is not a reason to slow down in a dangerous patient. It is a reason to choose the correct dangerous pathway. Ovarian torsion needs timely surgery to preserve ovarian function. Appendicitis needs surgical or imaging-directed management before perforation. Pyelonephritis with sepsis needs antibiotics and resuscitation. Ectopic rupture needs hemorrhage control. The resident should keep ectopic pregnancy high on the list, while still asking what other diagnosis would be harmed by delay.

The differential should be ordered by danger and reversibility. Ectopic rupture, ovarian torsion, appendicitis, sepsis, and major hemorrhage cannot be left waiting for

perfect certainty. A positive pregnancy test adds ectopic pregnancy to the urgent list and determines how pregnancy location is assessed, but it does not remove non-gynecologic disease.

Imaging decisions should follow maternal safety. Ultrasound is first-line for pregnancy location and pelvic assessment. But when the maternal diagnosis requires CT, MRI, radiography, or another modality, necessary imaging should not be withheld simply because the patient is pregnant. The Israeli imaging position paper states that no imaging modality is absolutely forbidden in pregnancy and that the risk from delayed diagnosis may exceed the risk from a properly indicated diagnostic examination. Clinically used single diagnostic examinations are below fetal-damage thresholds and are not by themselves an indication for pregnancy termination.

That point is practical in the emergency department. If appendicitis, bowel injury, renal obstruction, trauma, or another serious maternal diagnosis is plausible, the safest fetal decision may be correct maternal diagnosis. Maternal shock, sepsis, or missed surgical disease is not fetal protection. The resident should be able to explain that ultrasound answers one set of questions, and additional imaging may be needed to answer another.

There is also a communication discipline during emergencies. Tell the patient and family what is known, what is suspected, and why the tempo has changed. "We are worried about bleeding inside the abdomen." "We are preparing blood and the operating room because waiting could be dangerous." "The pregnancy location is not safe, and your symptoms suggest rupture." These sentences are clearer than vague reassurance and less frightening than unexplained rushing.

A useful bedside sequence is: assess stability, locate the pregnancy if time allows, look for blood, broaden the differential, and escalate when physiology demands it. In a stable patient, that sequence supports careful diagnosis. In an unstable patient, it compresses into resuscitation and source control. The resident's judgment is knowing when the second version has begun.

So this lecture should leave one practiced tempo in the listener's mind. Stable patient, careful localization. Unstable patient, resuscitation and source control. Symptomatic patient below the discriminatory zone, ultrasound anyway. Upper abdominal blood, escalate. A normal first hemoglobin, normal initial pressure, or temporarily improved pain cannot cancel a clinical picture of evolving intra-abdominal hemorrhage.